`Customer Segmentation & Churn Pattern Analytics in European Banking

Author: Sneha Raut
Project Mentor: Saiprasad Kagne

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Abstract

Customer churn remains one of the most significant yet under-addressed costs in European retail banking, eroding lifetime value, inflating acquisition expenses, and destabilizing revenue streams. Despite access to rich transactional and demographic data, most institutions still rely on aggregate chum metrics that fail to reveal granular risk patterns across customer segments, geographies, and financial profiles. This study presents a comprehensive customer segmentation and churn pattern analytics framework applied to a multi-country European banking dataset comprising 10.000 customer records from France, Spain, and Germany.
Employing a systematic, multi-dimensional segmentation approach-integrating geographic (country),demographic (age bands: <18,18-30,<30, 30-45, 46-60, 60+), behavioural (tenure groups, product ownership, activity status), and financial dimensions (credit-score bands, balance segments) the research quantifies overall and segment-specific churn rates while identifying high-risk cohorts, particularly among high-value customers. Advanced comparative analyses examine churn differentials by gender, geography-age interactions, and financial stability indicators. Key performance indicators developed in the study include segment churn rate, high-value churn ratio, geographic risk index, and engagement-drop indicator.
To translate analytical insights into actionable intelligence, a fully interactive Google Colab web application was developed, enabling dynamic filtering, drill-down visualizations, and real-time KPI monitoring for geography-wise, age-tenure, and high-value customer churn exploration. The framework demonstrates how segmentation-driven analytics can move chum management from generic, reactive strategies to precise, data-informed interventions. This research contributes both methodological rigor and practical decision-support tools to the field of banking analytics, offering banks a replicable blueprint for reducing churn, optimizing retention efforts, and safeguarding revenue stability across diverse European markets.

1. Introduction

1.1 Background

Customer churn represents one of the most significant challenges in retail banking. Research shows that acquiring a new customer cost five to seven times more than retaining an existing one, making churn prevention a top strategic priority for financial institutions. European banks operate in a highly competitive and increasingly digital landscape. Despite having rich customer-level data, many banks still rely on generic reactive retention approaches that fail to address root causes of churn in specific segments.

1.2 Problem Statement

Despite access to detailed customer data, European banks face three core challenges:

- *Inability to identify which specific customer groups are at highest risk of churning.*
- *Lack of understanding of how churn behaviour differs across geographies and demographics.*
- *Failure to quantify the financial impact of chum among high-value customer segments.*

1.3 Research Objectives

- *Measure the overall churn rate across the customer base.*
- *Identify chum distribution across key customer segments.*
- *Compare churn behaviour across European regions.*
- *Quantify the financial risk posed by high-value customer churn*
- *Evaluate the relationship between customer engagement and churn.*
- *Provide actionable recommendations for targeted retention strategies.*

2. Dataset Description

The dataset contains 10,000 European bank customers with 14 attributes covering demographics, financial profile, and banking behaviour. The target variable is Exited indicating whether the customer churned (1) or remained (0).

Column	Description	Type
Credit Score	Customer creditworthiness score	Numerical
Geography	Country: France, Spain, Germany	Categorical
Gender	Male/Female	Categorical
Age	Customer age in years	Numerical
Tenure	Years with the bank	Numerical
Balance	Account balance in EUR	Numerical
Num. Of Products	Number of bank products held	Numerical
HasCrCard	Credit card ownership (0/1)	Binary
IsActiveMember	Activity status (0/1)	Binary
Estimated Salary	Estimated annual salary in EUR	Numerical
Exited	Churn indicator - target variable (0/1)	Binary

Data quality assessment confirmed zero missing values across all columns, making the dataset immediately suitable for analysis without imputation.

3. Methodology

3.1 Data Ingestion, Validation, and Pre-processing

The dataset was loaded into a pandas Data Frame Validation checks included:

- **Completeness**: *No missing values were present across any column.*
- **Consistency**: *Binary variables (HasCrCard IsActive Member. Exited) were verified to contain only 0/1 values.*
- **Range validation**: *Credit Score (0-850), Age (18-92), Tenure (0-10), Balance (20), and Num Of Products (1-4) were confirmed to lie within realistic banking bounds.*
- **Duplicate detection**: *No duplicate Customer id entries existed.*

3.2 Cleaning and Feature Engineering

- *Non-analytical fields (Customerld, Surname) were dropped.*
- *Categorical variables (Geography, Gender) were label-encoded for grouping operations while preserving original strings for reporting.*
- *Derived segmentation features were created (detailed in Section 3.3).*
- *No outlier treatment was applied, as all values were business-valid (e.g., zero-balance accounts and age extremes are legitimate).*

3.3 Customer Segmentation Design

Segment	Column	Groups
Age Group	Age	Under 30/30-45/46-60/60+
Credit Band	Credit Score	Low (0-579)/Medium (580-669) /High (670-850)
Tenure Group	Tenure	New (0-2 yrs)/Mid (3-6 yrs)/Long (7+ yrs)
Balance Segment	Balance	Zero/Low/High (100k+)

4. Key Findings

4.1 Overall Churn Rate

The bank's overall chum rate stands at 20.37%, with 2,037 customers out of 10,000 having exited. Approximately 1 in 5 customers is leaving the bank.

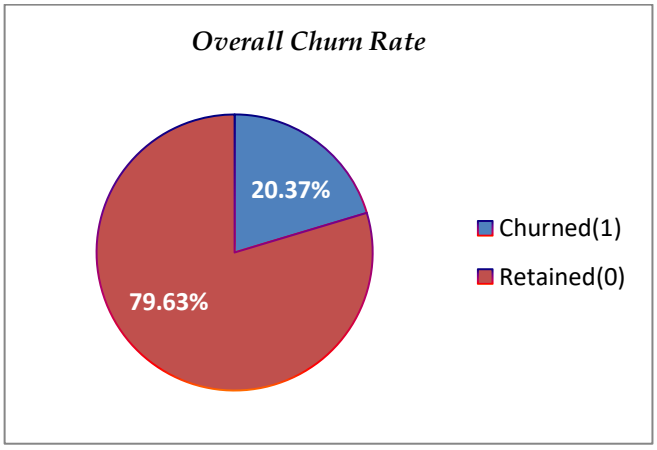

Figure 1: Overall Churn Distribution — 20.37% churned vs 79.63% reteamed

Row Labels	Sum of Customer count	Average of Percentage%
Churned(1)	2037	20.37%
Retained(0)	7963	79.63%
Grand Total	**10000**	**50.00%**

4.2 Geographic Analysis

Geography is the single strongest predictor of chum. Germany exhibits a churn rate of 32.44%-nearly double that of France (16.15%) and Spain (16.67%).

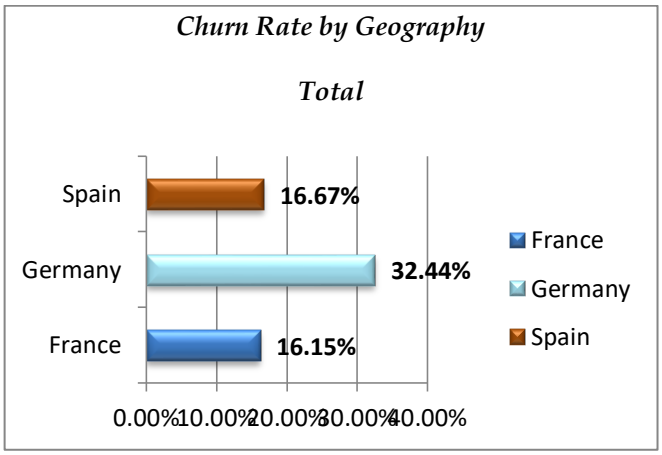

Figure 2: Churn Rate by Geography — Germany at 32.44% is the highest risk country

Churn Rate by Geography

Row Labels	Average of Exited
France	16.15%
Germany	32.44%
Spain	16.67%
Grand Total	**20.37%**

4.3 Gender Analysis

Female customers churn at a substantially higher rate. (25.07%) compared to male customers (16.46%), an 8.61 percentage point gap suggesting unmet needs among female customers.

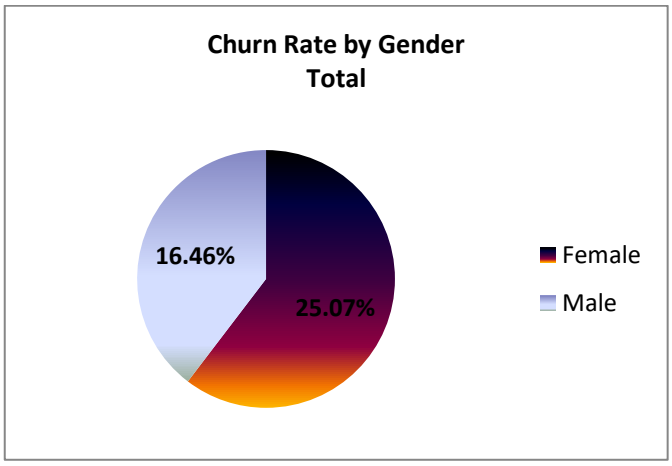

Figure 3: Churn Rate by Gender — Female customers Churn significantly more

Row Labels	Average of Exited
Female	25.07%
Male	16.46%
Grand Total	**20.37%**

4.4 Age Group Analysis

Customers aged 46 to 60 churn at a rate of 51.12% -- more than half of all customers in this age group are leaving the bank. This is 2.5 times the overall average.

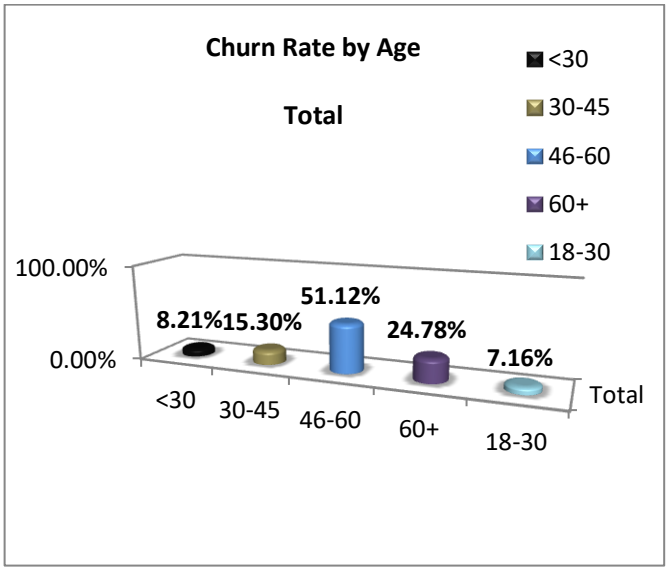

Figure 4: Churn Rate by Age Group — The 46-60 age group at 51.12% is critically at risk

Row Labels	Average of Exited
<30	8.21%
30-45	15.30%
46-60	51.12%
60+	24.78%
18-30	7.16%
Grand Total	**20.37%**

4.5 Balance Segment Analysis

Counter-intuitively, high-balance customers (above EUR 100,000) churn at the highest rate (25.23%), indicating that wealthy customers are actively seeking better services elsewhere.

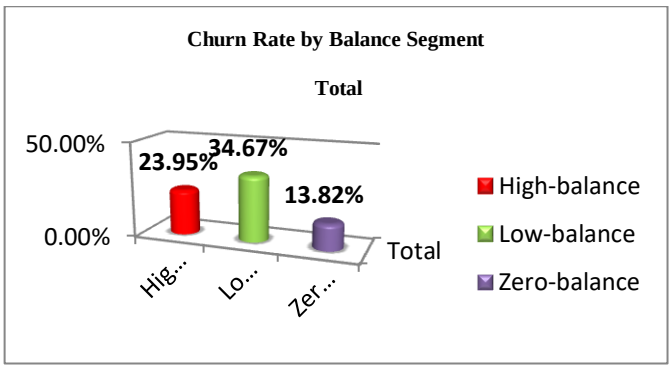

Figure 5: Churn Rate by Balance Segment — High balance customers Churn the most

Churn Rate by Balance Segment

Row Labels	Average of Exited
High-balance	23.95%
Low-balance	34.67%
Zero-balance	13.82%
Grand Total	**20.37%**

4.6 Number of Products Analysis

Customers with 2 products show the lowest churn rate at 7.58%. However, customers with 3 or 4 products show dramatically high churn rates of 82.71% and 100.00%, indicating over-selling drives dissatisfaction.

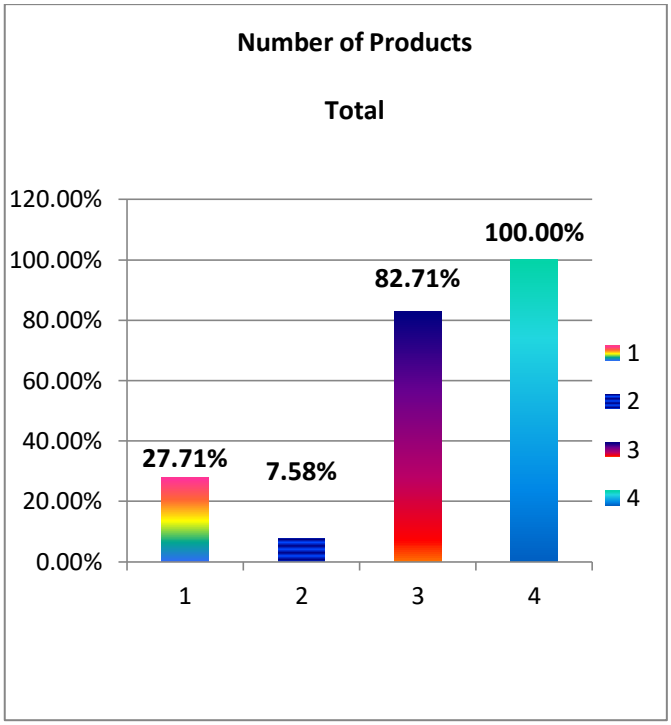

Figure 6: Churn Rate by Number of Products — 3.4 products Show extreme churn

Row Labels	Average of Churn Rate%
1	27.71%
2	7.58%
3	82.71%
4	100.00%
Grand Total	**54.50%**

4.7 Active Membership Analysis

Inactive customers churn at 26.85% compared to 14.27% for active members. With 48.49% of all customers currently inactive, this represents a major retention opportunity.

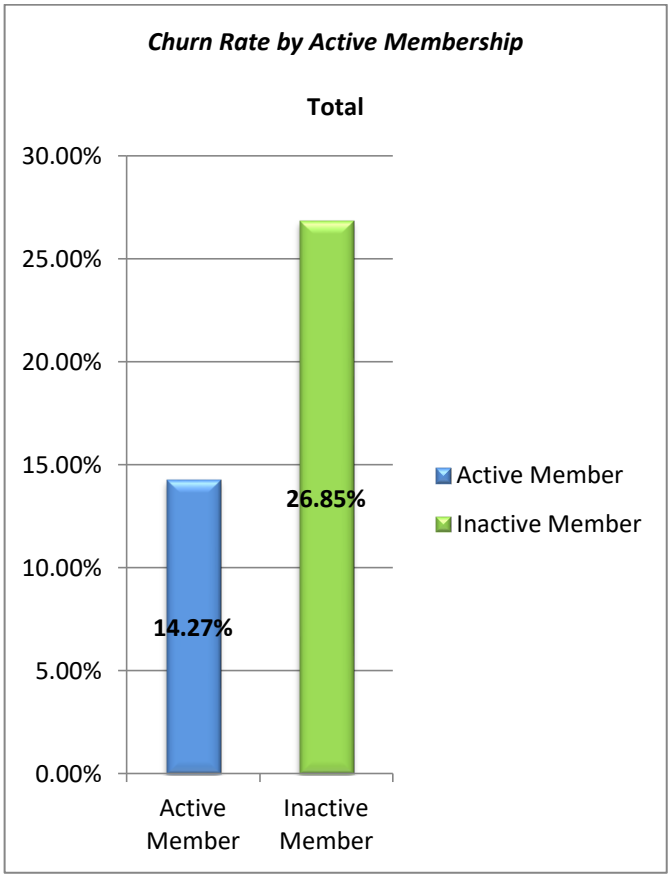

Figure 7: Churn Rate by Active Membership — Inactive members Churn at 1.9x the rate

Row Labels	Average of Exited
Active Member	14.27%
Inactive Member	26.85%
Grand Total	**20.37%**

4.8 Tenure Group Analysis

Tenure has a mild effect on churn, Newer customers. churn slightly more (21.15%) than long-term customers (19.51%), though the differences are small compared to other segments.

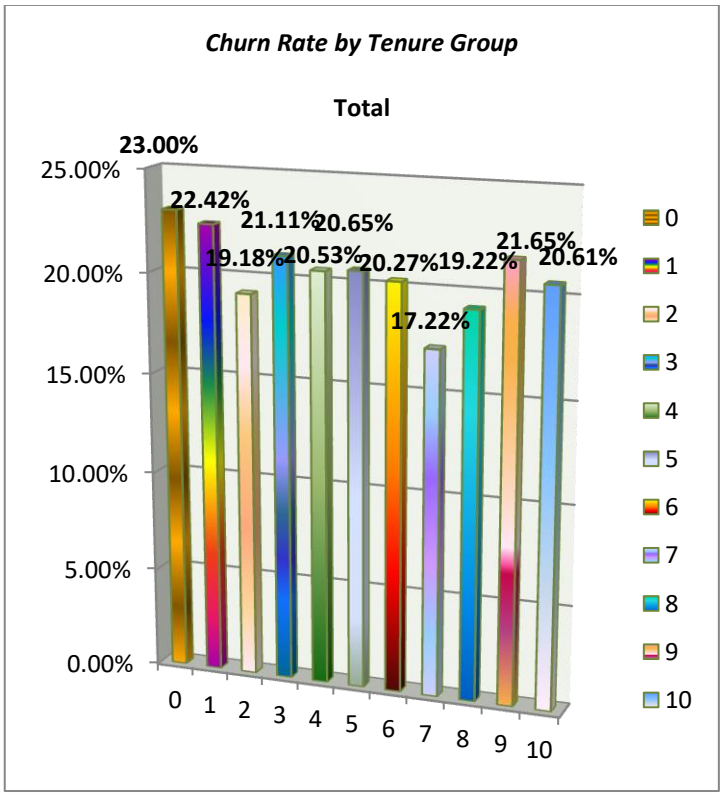

Figure 8: Churn Rate by Tenure Group — Minimal Difference across tenure segments

Row Labels	Average of Exited
0	23.00%
1	22.42%
2	19.18%
3	21.11%
4	20.53%
5	20.65%
6	20.27%
7	17.22%
8	19.22%
9	21.65%
10	20.61%
Grand Total	**20.37%**

4.9 Credit Band Analysis

Credit score has minimal impact on churn. Low credit band customers churn at 22.02% versus 19.32% for high credit band, suggesting credit score is not a strong chum predictor.

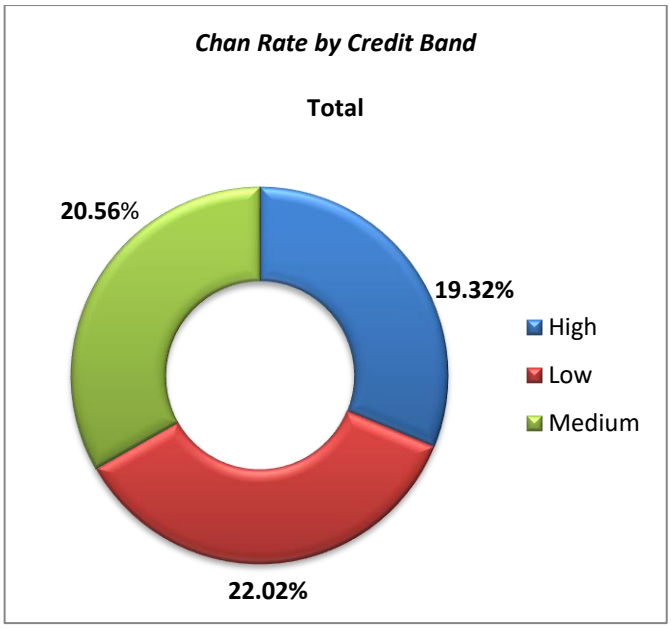

Figure 9: Chan Rate by Credit Band— Credit score shows minimal churn variation

Row Labels	Average of Exited
High	19.32%
Low	22.02%
Medium	20.56%
Grand Total	**20.37%**

5. Key Performance Indicators

KPI	Description	Value
Overall Churn Rate	% of total customers who exited	20.37%
High Value Churn Rate	Churn among balance > EUR 100k customers	25.23%
Geographic Risk Index	Germany churn vs bank average	1.59x
Engagement Gap	Inactive vs active member churn difference	12.58%
Balance at Risk	Total balance of churned high-value customers	EUR 159,489,691
Peak Age Churn Rate	Churn rate among 46-60 age group	51.12%
Most Loyal Segment	Customers with 2 products churn rate	7.58%
Highest Risk Segment	Germany 46-60 churn rate	67.33%

6. Strategic Recommendations

6.1 Priority 1 — Germany Retention Programme

Given Germany's churn rate of 32.44% and EUR 97.9M in balance at risk, an immediate country-specific retention programme is essential. The bank should conduct qualitative research to understand German customer dissatisfaction, introduce Germany-specific product offerings, and assign dedicated relationship managers to high-value German customers.

6.2 Priority 2- Middle-Aged Customer Engagement

Customers aged 46 to 60 represent the most critical at-risk segment with a 51.12% churn rate. The bank should develop tailored financial planning products for this life. stage including retirement planning, wealth management, and mortgage refinancing. Proactive

outreach via relationship managers is recommended for all customers in this segment.

6.3 Priority 3- Inactive Member Re-Engagement

With nearly half of all customers inactive and inactive members churning at 1.9 times the rate of active ones, a structured re-engagement campaign is critical. The bank should implement automated engagement triggers, personalised incentives, and regular check-in calls from relationship managers.

6.4 Priority 4-High Value Customer Retention

The EUR 159.5M balance at risk demands a premium retention strategy. The bank should create a dedicated VIP banking tier with exclusive benefits, personalised wealth management, and zero-wait customer service for customers with balances above EUR 100,000.

6.5 Priority 5- Product Portfolio Optimisation

The extreme churn rates among customers with 3 or 4 products strongly suggest over-selling is driving dissatisfaction. The bank should immediately audit sales practices, review product bundling strategies, and focus on quality of product fit rather than quantity.

6.6 Priority 6 - Gender-Inclusive Banking

The 8.6 percentage point churn gap between female (25.07%) and male (16.46%) customers indicates unmet needs. The bank should conduct gender-specific research, develop targeted products, and ensure female representation in product design teams.

7. Conclusion

This study demonstrates that customer chum in European retail banking is not a uniform phenomenon but a highly segmented risk that varies significantly across geographic, demographic, behavioural, and financial dimensions. Through systematic multi-dimensional segmentation and detailed pattern analysis on a 10,000-customer dataset spanning France, Spain, and Germany, the research successfully identified distinct high-risk customer cohorts that remain hidden when using traditional aggregate churn metrics.

Key findings reveal that chum is disproportionately concentrated among specific segments particularly younger customers, certain geographic markets, low-engagement clients, and paradoxically, a notable portion of high-value customers. The analysis highlights critical interaction effects, such as geography-age and financial stability-churn relationships, providing banks with actionable intelligence to move beyond generic retention campaigns toward precise, segment-targeted interventions.

This research has delivered a comprehensive, segmentation-driven analysis of customer churn across European banking. By systematically examining churn patterns across geography, demographics, financial profile, and engagement, several critical insights have emerged.

The findings demonstrate that churn is not uniformly distributed. Specific segments particularly German customers, those aged 46 to 60, inactive members, and high-balance customers face dramatically elevated churn risks. Over EUR 159 million in customer balances is at risk.

The six strategic recommendations provide a clear, prioritised roadmap for retention interventions. By targeting the highest-risk segments with tailored strategies, the bank can significantly reduce churn rates, protect revenue, and improve long-term customer lifetime value.

The development and deployment of an interactive Streamlit web application further bridges the gap between advanced analytics and practical decision-making. Banking executives and retention teams can now dynamically explore churn patterns, monitor real-time KPIs, and quantify revenue risk from high-value

customer attrition transforming churn management from reactive to proactive and data-driven.

8. References

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